
Investigating the Impact of Different Loss Functions on Regression Tasks

Objective

The goal is to explore how different loss functions affect the convergence speed, model performance, and robustness to outliers in regression tasks. This lab will deepen your understanding of the strengths and limitations of various loss functions and their practical applications.

Instructions

1. **Dataset Preparation:**
 - Use a suitable regression dataset (e.g., Boston Housing or a synthetic dataset).
 - Perform necessary preprocessing like scaling or encoding where applicable.
2. **Implementation:**
 - Use Python and relevant libraries (e.g., NumPy, Scikit-learn, Matplotlib, PyTorch).
 - Implement linear regression models for tasks, comparing different loss functions.
3. **Evaluation and Analysis:**
 - Analyze results, document observations, and draw conclusions.
 - Answer questions based on experimental findings.

Task 1: Effect of Loss Functions on Convergence Speed

Question: How do different loss functions influence the convergence speed of a regression model?

1. **Setup:**
 - Load and preprocess a regression dataset (e.g., Boston Housing or a synthetic dataset).
 - Implement a linear regression model.
2. **Loss Functions to Compare:**
 - Mean Squared Error (MSE)
 - Mean Absolute Error (MAE)
 - Huber Loss
3. **Experimentation:**
 - Train the model using Gradient Descent with each loss function.
 - Track the number of epochs needed to reach a specific loss threshold (e.g., $1e-3$).
4. **Deliverables:**
 - Plot loss vs. epochs for each loss function.
 - Identify which loss function converges faster and discuss why.

Task 2: Robustness to Outliers

1. **Deliverables:**
 - Plot predicted vs. actual values for each loss function.
 - Analyze which loss function is most robust to outliers and explain why.

Visualization and Analysis

Plots

- **Task 1:** Loss vs. epochs for different loss functions.
- **Task 2:** Predicted vs. actual values for each loss function.

Submission Requirements

Code: Submit clean and well-commented Python code for all tasks.

Report: a) Include labeled plots and detailed explanations.

b) Answer the posed questions comprehensively.